LINE SEGMENTS AND INTERCEPTS

ANALYTIC GEOMETRY

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GRADING SYSTEM

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30%

PRELIM

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EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Line Segments

- Distance Formula
- Division of Line Segment
- Midpoint of Line Segment

Intercepts of the Graph of an Equation

- x-intercepts
- y-intercepts

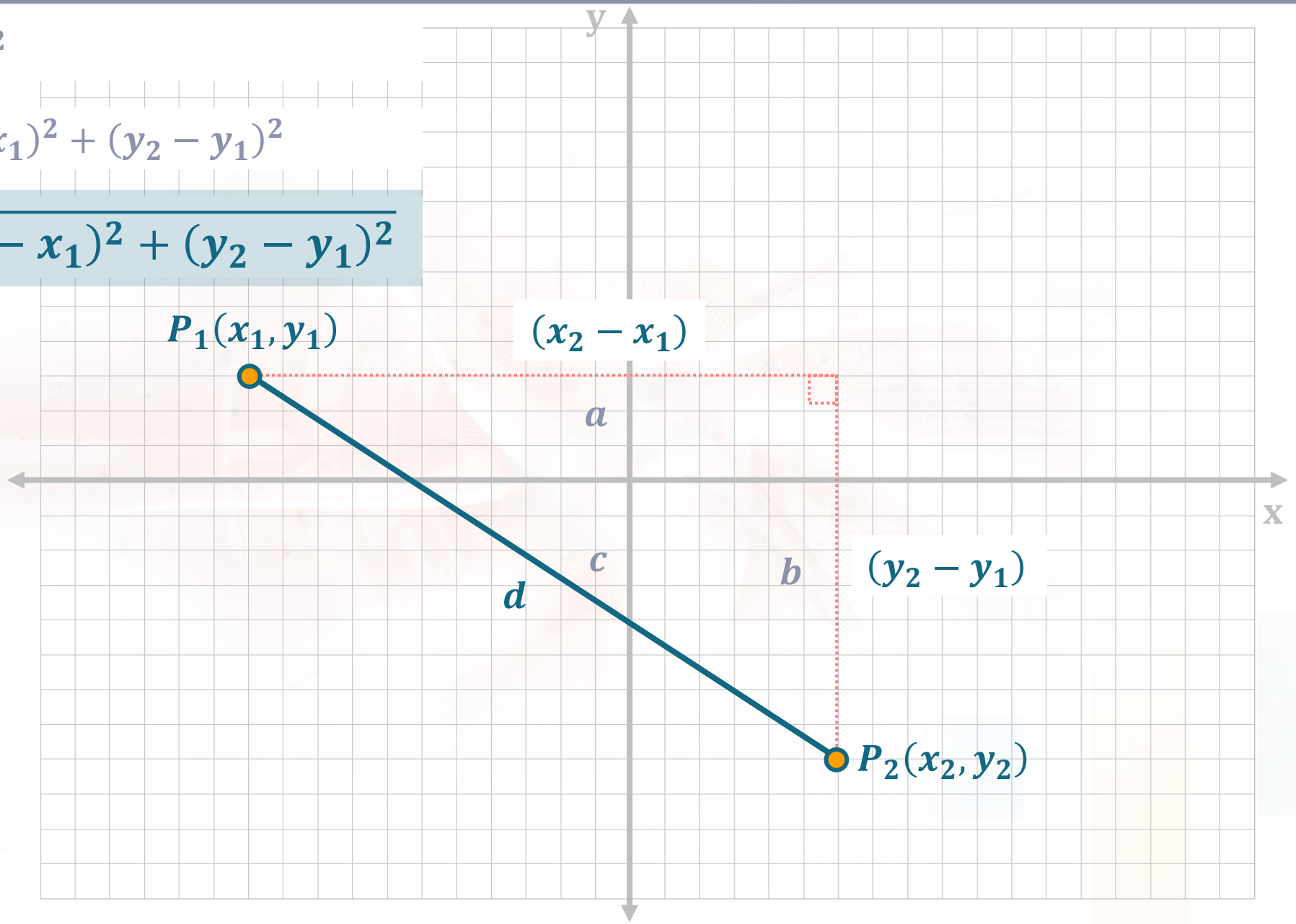
LINE SEGMENTS

THE DISTANCE FORMULA

$$c^2 = a^2 + b^2$$

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\boldsymbol{d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$



Find the **distance** between the points $(4, -2)$ and $(6, 5)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

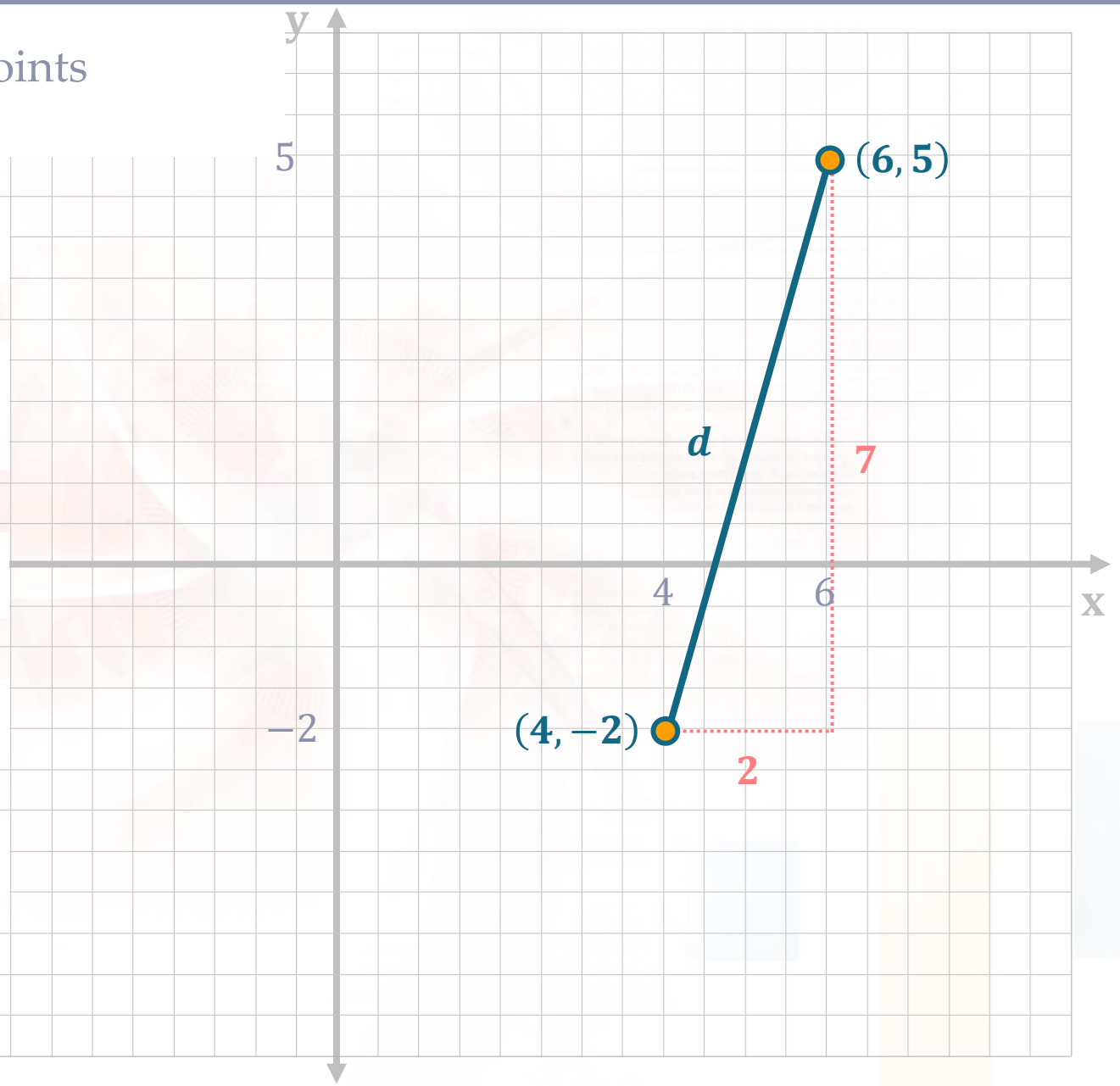
$$d = \sqrt{(6-4)^2 + \left(5 - (-2)\right)^2}$$

$$d = \sqrt{2^2 + 7^2}$$

$$d = \sqrt{53}$$

$$\boxed{d = 7.28 \text{ units}}$$

ans

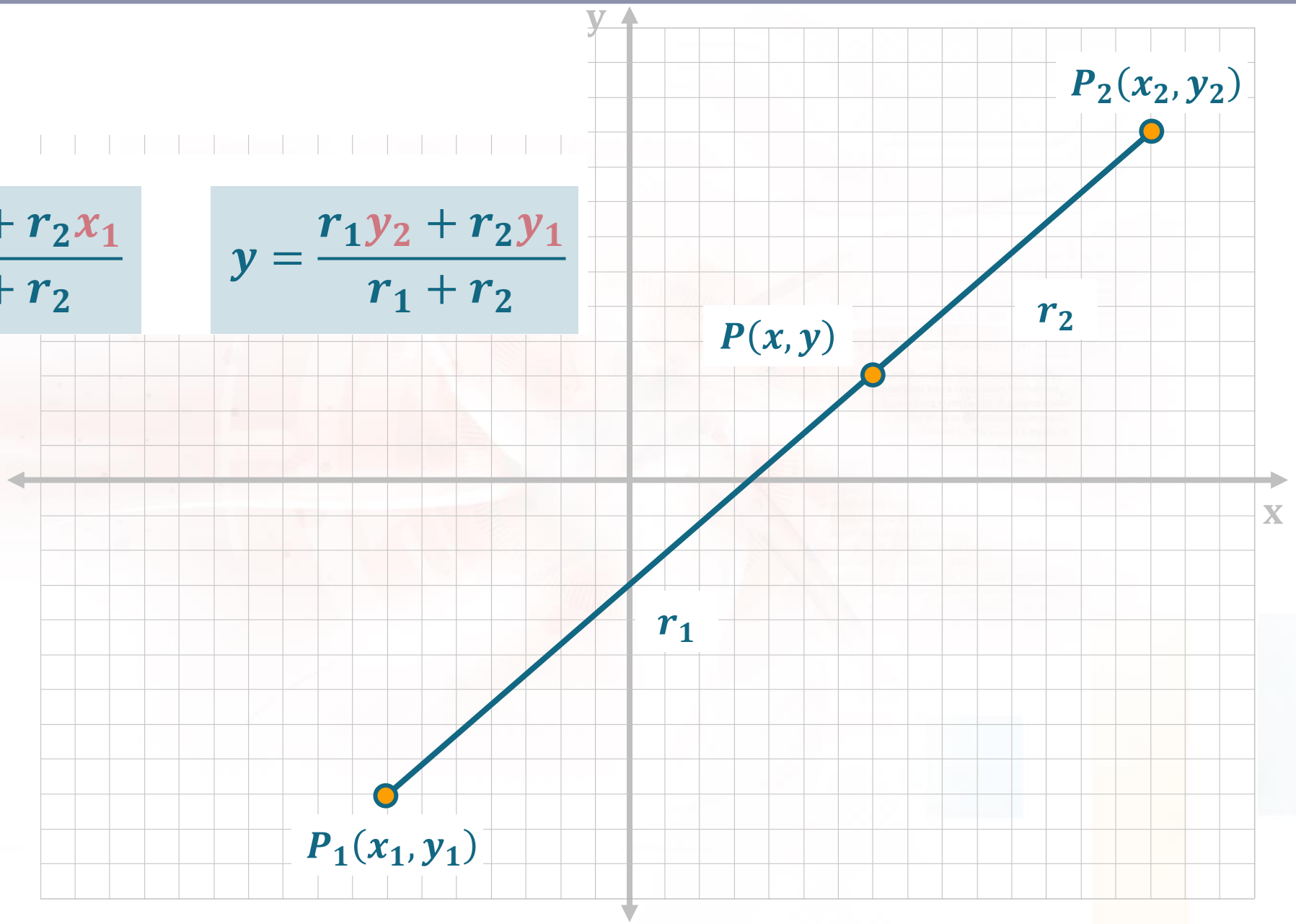


THE DIVISION OF LINE SEGMENT

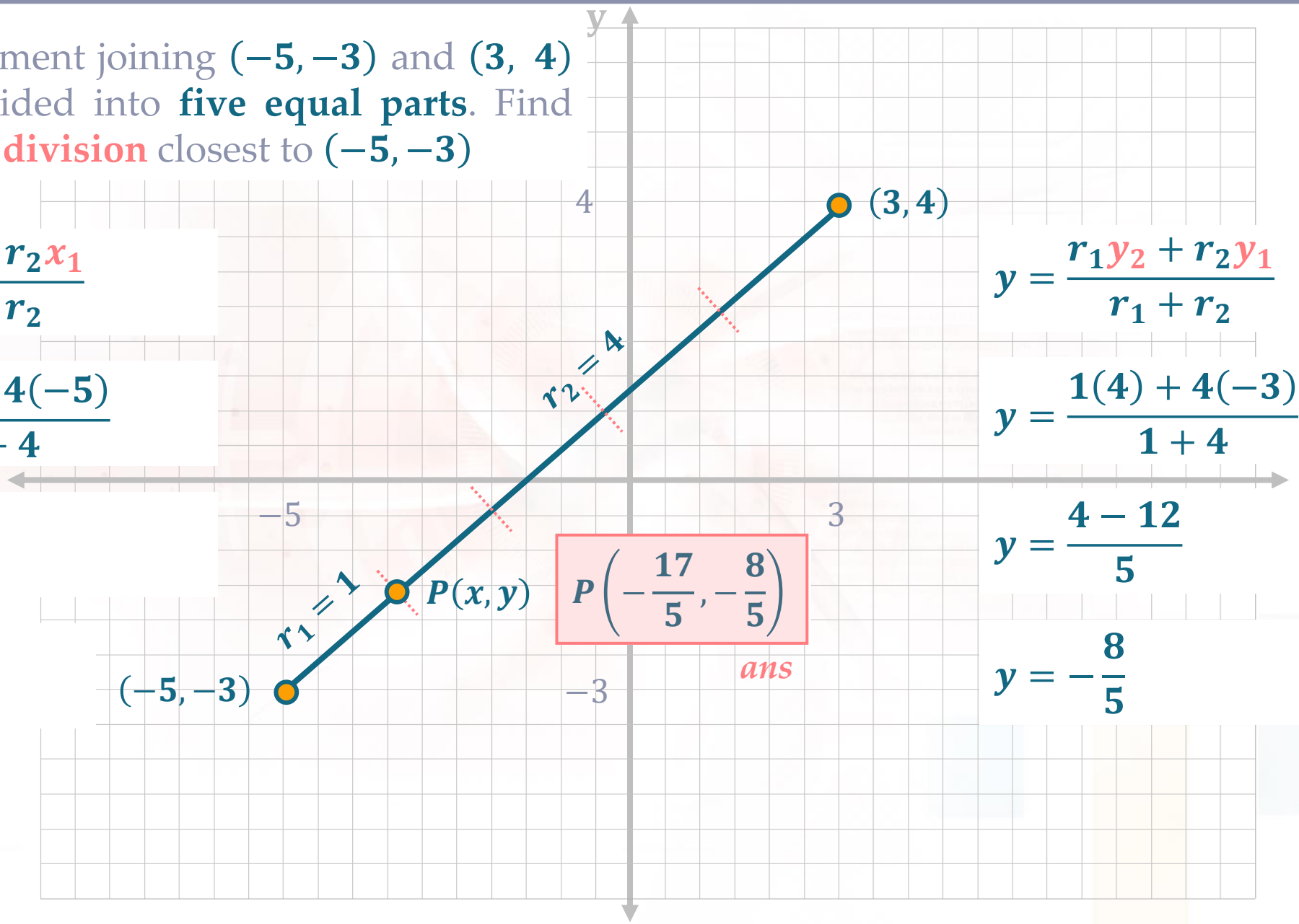
$$\frac{P_1P}{PP_2} = \frac{r_1}{r_2}$$

$$x = \frac{r_1x_2 + r_2x_1}{r_1 + r_2}$$

$$y = \frac{r_1y_2 + r_2y_1}{r_1 + r_2}$$



The line segment joining $(-5,-3)$ and $(3,\ 4)$ is to be divided into **five equal parts**. Find the **point of division** closest to $(-5,-3)$



$$P\left(-\frac{17}{5}, -\frac{8}{5}\right)$$

ans

$$x = \frac{r_1 x_2 + r_2 x_1}{r_1 + r_2}$$

$$x = \frac{1(3) + 4(-5)}{1 + 4}$$

$$x = \frac{3 - 20}{5}$$

$$x = -\frac{17}{5}$$

$$y = \frac{r_1 y_2 + r_2 y_1}{r_1 + r_2}$$

$$y = \frac{1(4) + 4(-3)}{1 + 4}$$

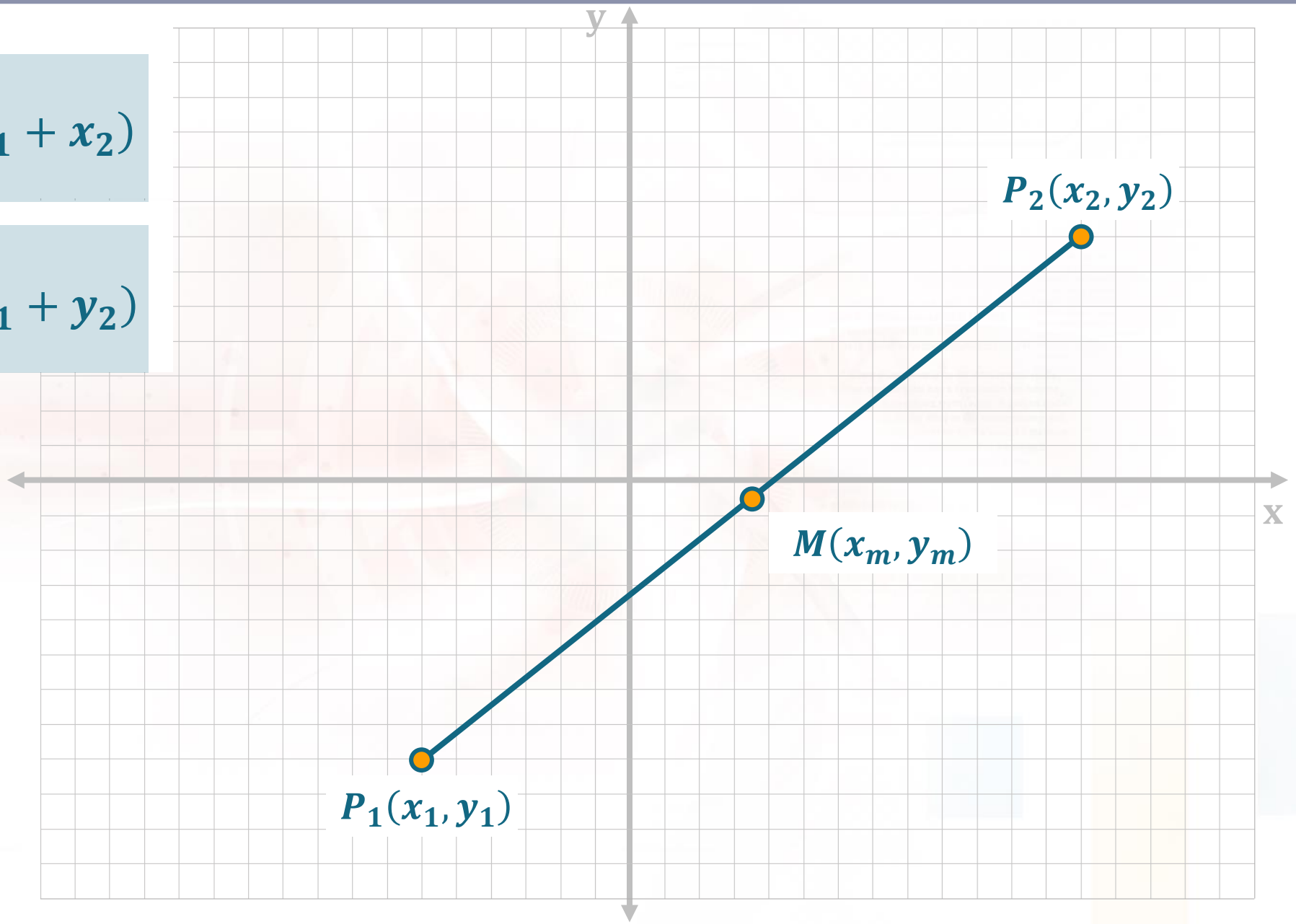
$$y = \frac{4 - 12}{5}$$

$$y = -\frac{8}{5}$$

MIDPOINT OF A LINE SEGMENT

$$x_m = \frac{1}{2}(x_1 + x_2)$$

$$y_m = \frac{1}{2}(y_1 + y_2)$$



Find the **midpoint** of the segment joining $(7, -2)$ and $(-3, 5)$

$$x_m = \frac{1}{2}(x_1 + x_2)$$

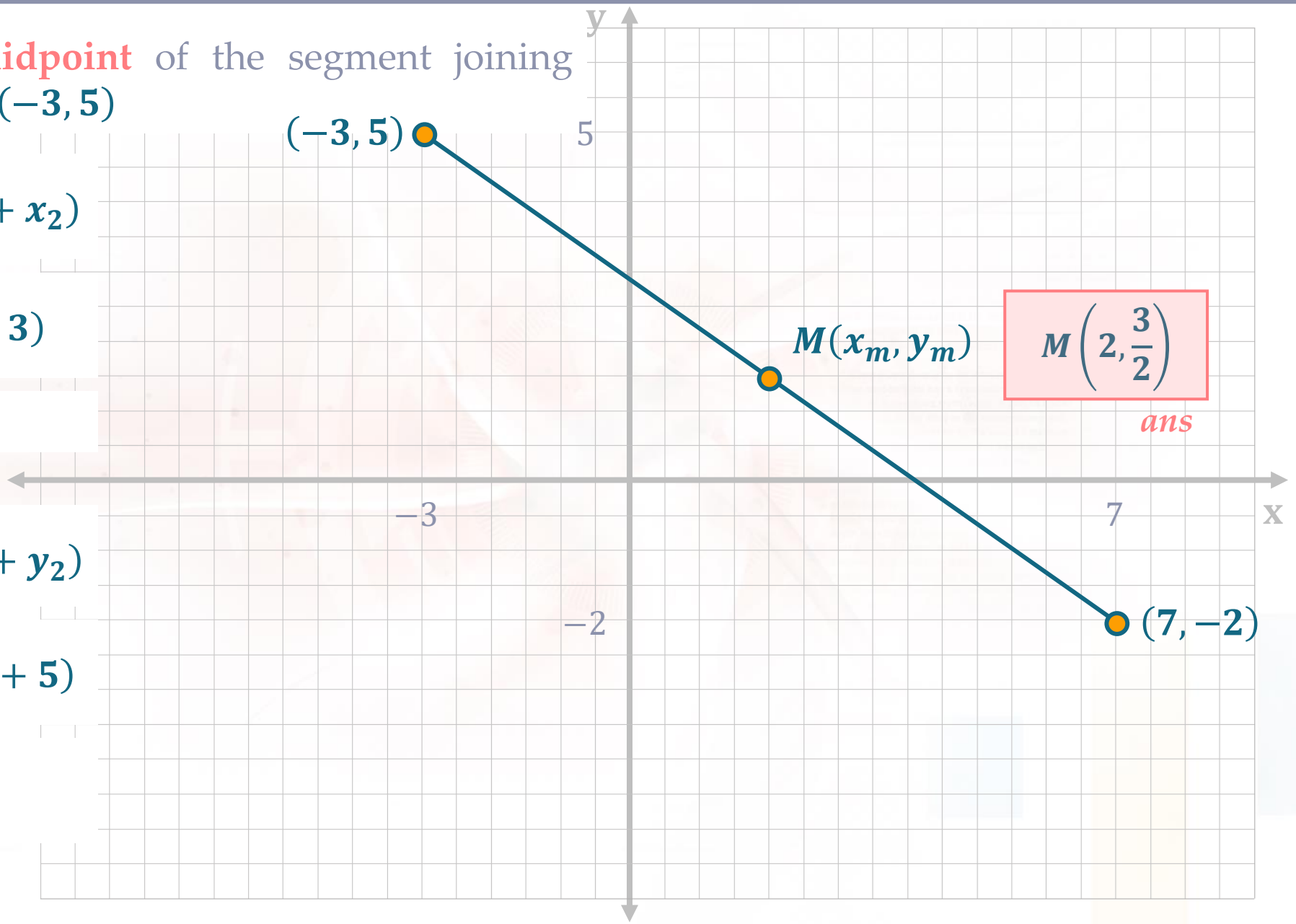
$$x_m = \frac{1}{2}(7 - 3)$$

$$x_m = 2$$

$$y_m = \frac{1}{2}(y_1 + y_2)$$

$$y_m = \frac{1}{2}(-2 + 5)$$

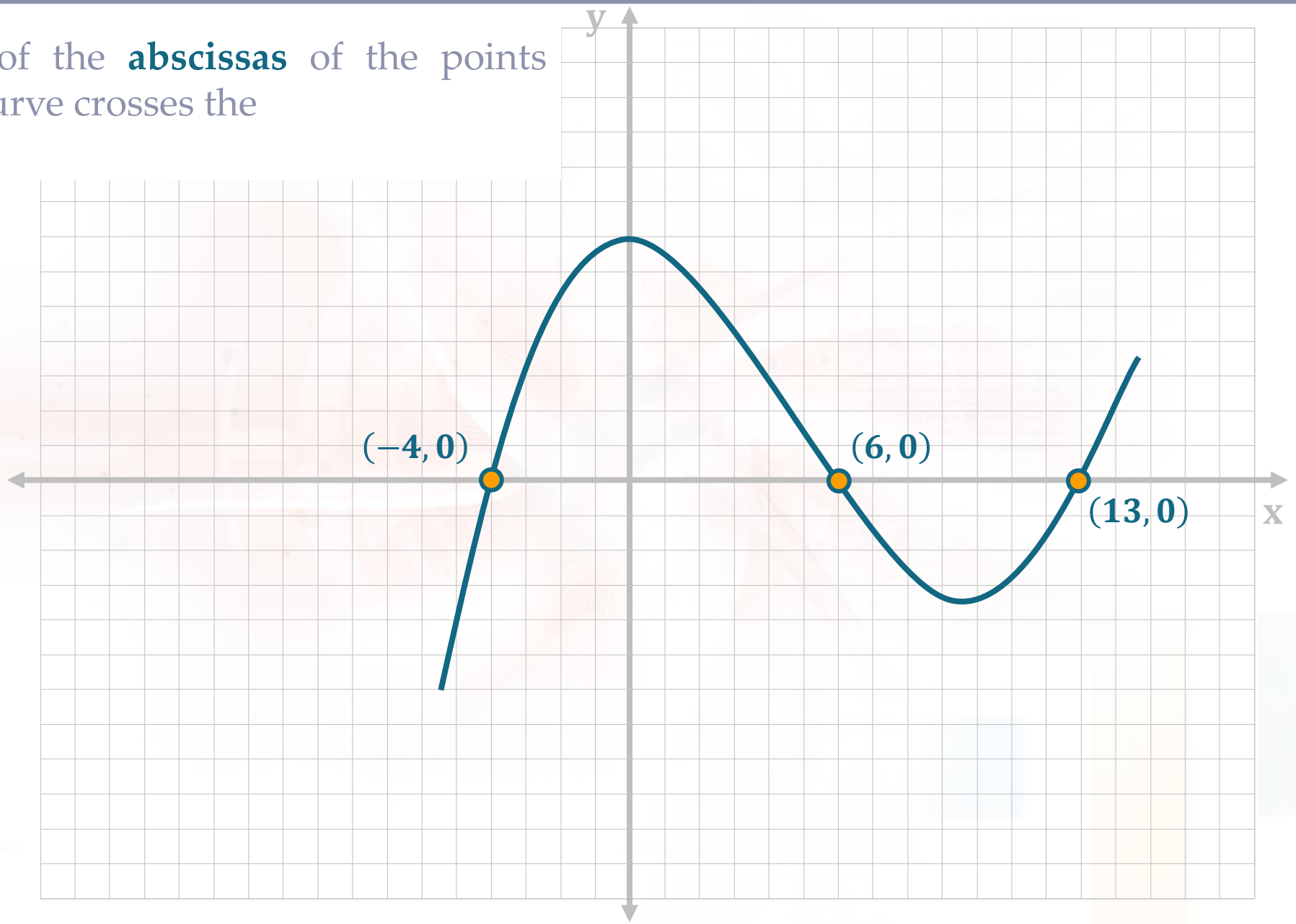
$$y_m = \frac{3}{2}$$



INTERCEPTS OF THE GRAPH OF AN EQUATION

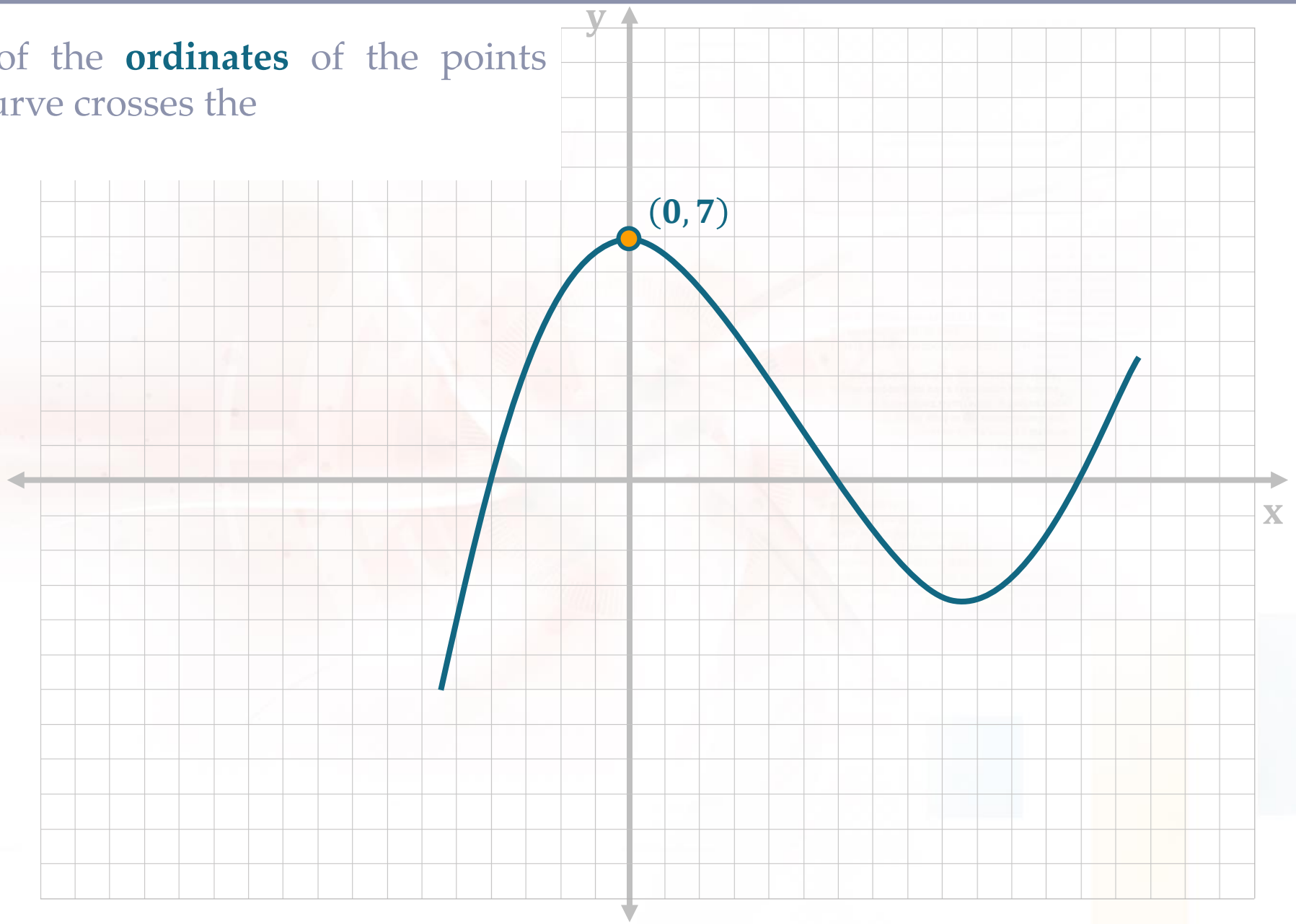
x - INTERCEPTS

the values of the **abscissas** of the points where the curve crosses the x-axis

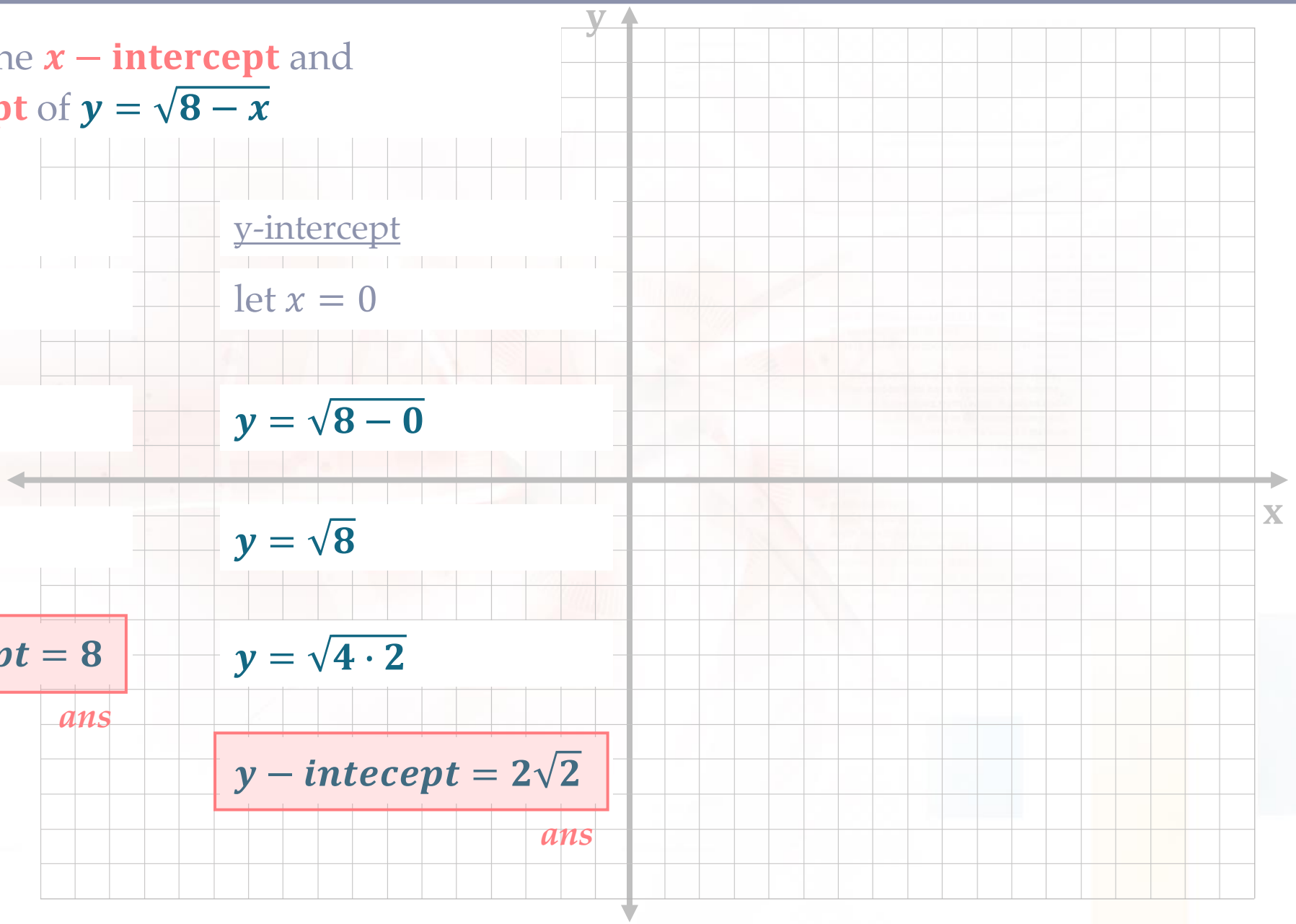


y - INTERCEPTS

the values of the **ordinates** of the points where the curve crosses the y-axis



Determine the $x-$ **intercept** and $y-$ **intercept** of $y=\sqrt{8-x}$



x-intercept

let $y=0$

$0=\sqrt{8-x}$

$0=8-x$

$x-intecept=8$

ans

y-intercept

let $x=0$

$y=\sqrt{8-0}$

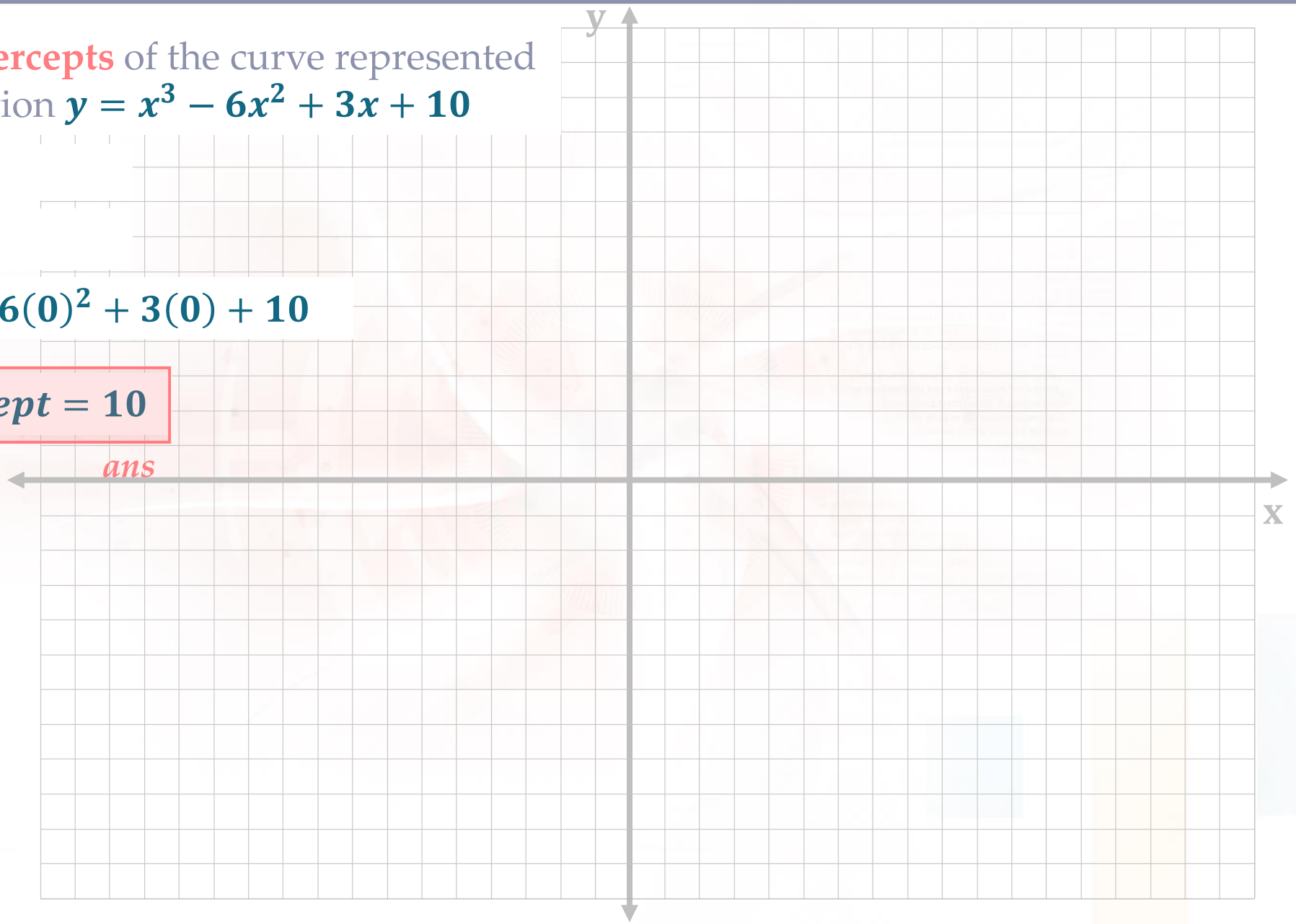
$y=\sqrt{8}$

$y=\sqrt{4\cdot 2}$

$y-intecept=2\sqrt{2}$

ans

Find the **intercepts** of the curve represented by the equation $\boldsymbol{y = x^3 - 6x^2 + 3x + 10}$



y-intercept

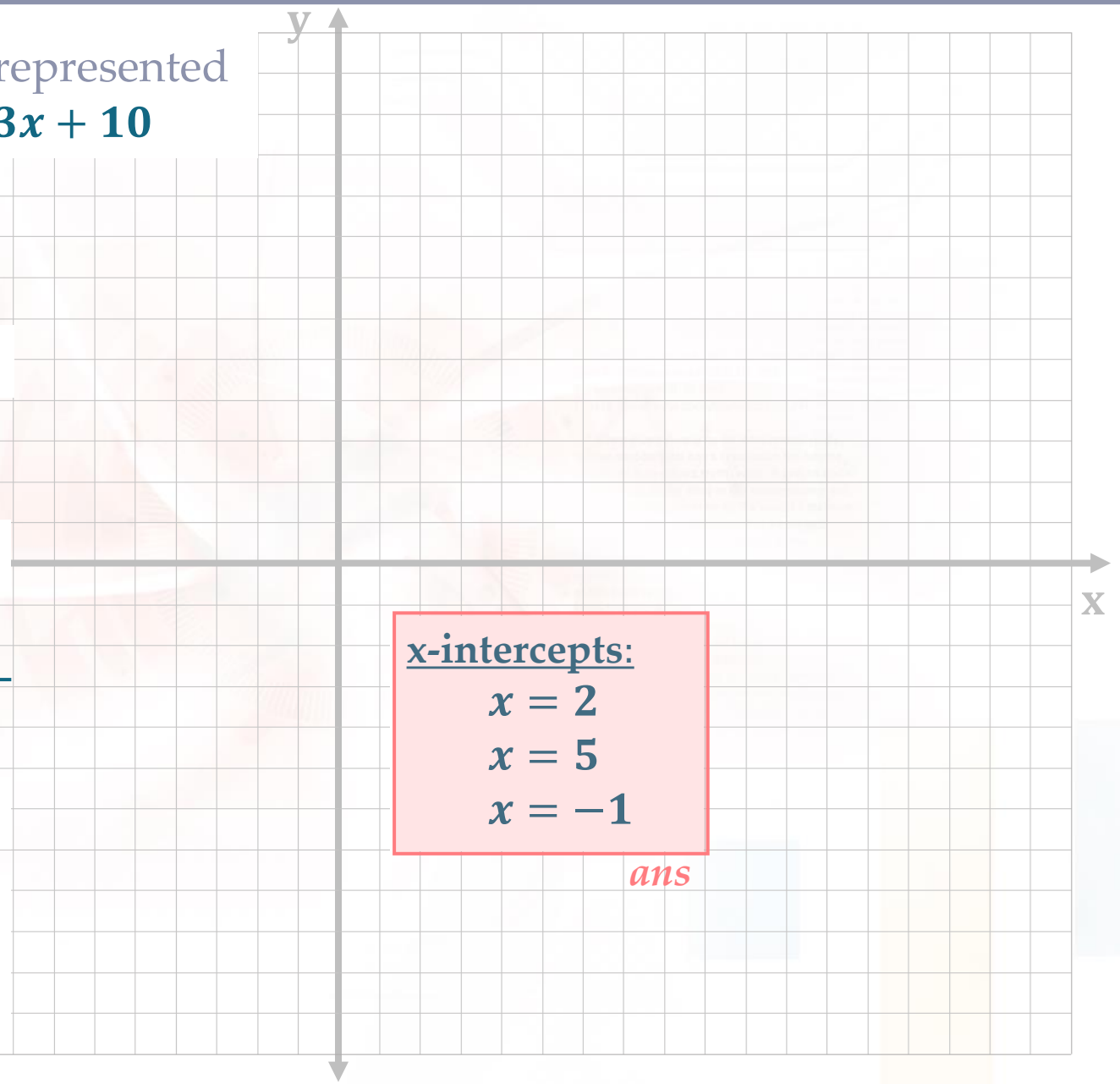
let $x = 0$

$$\boldsymbol{y = (0)^3 - 6(0)^2 + 3(0) + 10}$$

$$\boxed{\boldsymbol{y - intercept = 10}}$$

ans

Find the **intercepts** of the curve represented by the equation $y = x^3 - 6x^2 + 3x + 10$



x-intercept

let $y = 0$

$$0 = x^3 - 6x^2 + 3x + 10$$

Synthetic Division

$x = 2,$	1	-6	3	10
		2	-8	-10
	1	-4	-5	0

$$x^2 - 4x - 5 = 0$$

$$(x - 5)(x + 1) = 0$$

$$x = 5 \qquad x = -1$$

x-intercepts:

$x = 2$

$x = 5$

$x = -1$

ans

SEATWORK